



FINAL YEAR PROJECT STUDENT LOGBOOK

PROJECT TITLE

myMove: IoT Enabled Smart Parking Reservation and Privacy Vehicle Communication System

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SECTION A : LOGBOOK TYPE

LOGBOOK	:		FYP 1	✓	FYP 2
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SECTION B : STUDENT INFORMATION

NAME : SAABIRESH LETCHUMANAN

PROGRAM : BCS

STUDENT ID : B02240005

SEMESTER : MAY 2026

SECTION C : SUPERVISOR INFORMATION

MAIN

SUPERVISOR : MUHAMMAD SYAHMIE SHABARUDIN

CO-

SUPERVISOR :



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK NOTES

1. This logbook needs to be used by FYP1 and FYP2 students for the purpose of reporting all progress made in the project.
2. It is the responsibility of the student to keep the logbook updated. All suggestions, ideas, comments and instructions given by the supervisor need to be recorded in the logbook and to be complied in the project.
3. It is compulsory for the student to have a regular meeting with his/her supervisor. **A minimum of 6 meeting is required for each FYP.**
4. The FYP Committee and / or the supervisor have the right not to allow or not to recommend the student to present his work in FYP Presentation Week if student's progress does not achieve the minimum requirement set for FYP.
5. This logbook must be submitted together with the FYP report as stated in the Final Year Project Handbook.
6. The FYP Committee reserves right not to accept thesis/report for examination if this logbook is not properly documented.



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 21 / 05 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The outcomes from FYP 1 were reviewed, including the requirements specification, system architecture, pilot survey findings, and UI wireframes. The FYP 2 development workspace was then initialized with the Flutter SDK, Firebase BaaS project, and Node.js environment. The necessary IoT hardware components, including ESP32 microcontrollers, HC-SR04 ultrasonic distance sensors, and a breadboard, were also procured. Updates regarding the FYP 2 course schedule and briefing from Madam Faiqah on the thesis templates and milestone requirements were reviewed. Finally, the FYP 2 milestone roadmap and timeline breakdown were presented.
Supervisor (Suggestion & Comments)	The transition from the design phase to the full implementation phase was approved. It was emphasized that although the technical development is progressing smoothly, academic thesis writing is equally important, as it contributes significantly to the overall grading and should not be neglected. Guidance was provided to follow a six-chapter thesis structure. The use of a dummy payment integration for academic prototyping was also validated.
Next Meeting Plan	The draft of Chapters 1, 2, and 3 was compiled and submitted to the supervisor for structure and content review. Development of the UI modules and hardware integration was also initiated.
Supervisor Signature	



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 08 / 06 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The working User Authentication system, Firebase Firestore database connection, configured parking locations, GPS-based distance calculation, and dynamic Date & Time Picker with hourly pricing were successfully demonstrated. The interactive 2D parking layout was showcased with real-time slot availability, booking lifecycle management, countdown timers, and parking extension functionality with updated payment totals. The initial user interface for the Double-Parking QR Code privacy feature was also demonstrated.
Supervisor (Suggestion & Comments)	Impressed with the overall UI and features that are already developed
Next Meeting Plan	The remaining UI modules were continued alongside the preparation of the IoT hardware components. Documentation was also continued and will be presented during the next meeting.
Supervisor Signature	



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 25 / 06 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The initial thesis draft for Chapters 1–3 was presented and discussed, focusing on chapter restructuring, project objectives, and methodology alignment. The IoT hardware and Firebase Realtime Database telemetry were demonstrated using the ESP32 and HC-SR04 ultrasonic sensor, along with the Privacy Vehicle Communication module, zero-install Web Portal, push notifications, and centralized React-based Admin Portal. The tabletop physical prototype demonstration strategy for the upcoming final defense was also discussed.
Supervisor (Suggestion & Comments)	The project objectives and thesis structure were refined, including improvements to Objectives 2 and 3 and the organization of Chapters 2–6. The Zero-Install QR Web Portal and Admin Web Portal were highlighted as strong features, with support provided for the miniature parking lot prototype and 3D-printed components. The final defense timeline was also discussed, with development and documentation to be completed by Week 13.
Next Meeting Plan	Refine Chapters 1–3, draft Chapter 4 covering the system development and implementation, and finalize the hardware enclosure and tabletop diorama setup. Supervisor will review the revised chapters and Chapter 4 draft, as well as coordinate access to the 3D printing lab and miniature vehicle assets for the final prototype defense.
Supervisor Signature	



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 20 / 07 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The 3D-printed parking lot diorama and sensor placement were discussed, including the custom mounting design for the HC-SR04 sensors. The MapTiler integration, multi-sensor IoT demonstration, and comprehensive Admin Web Portal features were showcased, including real-time parking updates, location management, bookings, notifications, feedback, and SOS alerts. The Zero-Install Web Scanner with anti-spam geofencing was also demonstrated, alongside progress on Chapters 4, 5, and 6 of the thesis.
Supervisor (Suggestion& Comments)	The supervisor scheduled the 3D printing of the diorama using modular parts to simplify wiring, sensor mounting, and future adjustments, and offered miniature toy cars with printed QR codes for the final demonstration. The 100-meter GPS anti-spam mechanism, photo verification, MapTiler integration, and multi-tier Admin Portal were praised for their technical and practical value. Was advised to submit Chapters 4–6 for review, ensure functional and non-functional requirements are properly documented, and keep the weekly logbook updated for supervisor endorsement.
Next Meeting Plan	Will collect and assemble the 3D-printed diorama, install the ESP32 sensors and wiring, attach QR stickers to the toy cars, and test the complete scan-to-web-chat flow. The student will also complete and submit Chapters 4–6, continue updating the weekly logbook, and prepare the prototype for final testing. The supervisor will assist with the 3D printing process, provide miniature toy cars, and review the upcoming Chapters 4–6 thesis draft.
Supervisor Signature	



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 06 / 08 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The draft of Chapter 4 was submitted, covering the Flutter mobile app, ESP32 firmware, Firebase backend, React Admin Portal, and core implementation code. The thesis structure, diagrams, database tables, and architecture were reviewed with the supervisor, while the newly 3D-printed modular parking lot prototype was inspected for dimensions and sensor mounting. Preparations for the final FYP presentation and potential Innovation Showcase nomination were also discussed.
Supervisor (Suggestion & Comments)	The supervisor advised improving Chapter 4 by adding explanations for each code snippet, explicitly referencing all figures in the text, and correctly formatting database schemas and data dictionaries as tables. The 3D-printed diorama components were scheduled for collection and assembly with the hardware, while Chapters 5 and 6 were planned for review the following week. The supervisor also emphasized polished presentation and a seamless hardware-software demonstration to strengthen the project's evaluation during the final viva.
Next Meeting Plan	The student will revise Chapter 4 based on the supervisor's feedback, assemble the 3D-printed diorama with the ESP32 sensors and wiring, and finalize Chapters 5 and 6 for submission. The supervisor will complete the remaining 3D printing and hand over all diorama components to the student. The supervisor will also review the submitted Chapters 5 and 6 drafts.
Supervisor Signature	



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 09 / 08 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The initial Innoverse exhibition poster was presented, covering the problem statement, system architecture, methodology, and core system interfaces. The thesis documentation was reported to be near completion and ready for final submission review. The status of the completed 3D-printed parking diorama components was also discussed for final wiring and sensor mounting.
Supervisor (Suggestion& Comments)	The supervisor provided feedback to improve the Innoverse poster by focusing on evaluation results, chart readability, high-quality visual assets, consistent colors, larger UI screenshots, and clearer technology indicators in the methodology flowchart. The supervisor confirmed that all 3D-printed diorama components were completed and ready for collection, assembly, and wiring. After reviewing the revised poster, formal approval was granted for the official Innoverse showcase submission.
Next Meeting Plan	Will collect and assemble the 3D-printed diorama, mount the ESP32 sensors, finalize and submit the complete Chapters 1–6 thesis, and prepare the final presentation and prototype demonstration. The supervisor will hand over the completed diorama and conduct a final comprehensive review and grading verification of the thesis.
Supervisor Signature	

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 10 / 08 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 /
Student (Meeting Minute/ Achievements/ Activities)	The finalized 3D-printed modular parking diorama was collected and inspected, including the sensor mounting brackets and wiring channels for the ESP32 and HC-SR04 sensors. The complete FYP 2 thesis report covering Chapters 1–6 was officially submitted for comprehensive supervisor feedback, formatting verification, and grading review.
Supervisor (Suggestion & Comments)	The supervisor advised securing all sensor jumper wires beneath the diorama base to prevent loose connections during the live demonstration and recommended conducting multiple end-to-end dry runs to ensure reliable hardware, Firebase, mobile app, and web portal synchronization. The complete thesis draft was also acknowledged and accepted for final academic review, formatting verification against UNIMY guidelines, and grading assessment.
Next Meeting Plan	Will complete the physical diorama assembly, conduct a full end-to-end rehearsal of the hardware and software components. The supervisor will review the complete thesis report, provide final corrections and remarks, and prepare it for final endorsement.
Supervisor Signature	

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 13 / 08 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9
Student (Meeting Minute/ Achievements/ Activities)	The completion and submission of the FYP 2 thesis report was confirmed. The initial hardware prototype was demonstrated on the 3D-printed parking diorama, and the use of temporary double-sided tape for sensor alignment was discussed as unsuitable for the final defense. A customized HC-SR04 sensor mounting bracket design was presented, with a request to 3D-print three units using the lab's printer for a cleaner and more professional prototype.
Supervisor (Suggestion & Comments)	The supervisor approved the sensor bracket design after confirming its compatibility with the HC-SR04 sensors and diorama slots, with three brackets approved for 3D printing to improve cable management and sensor stability.
Next Meeting Plan	Will collect and install the 3D-printed sensor mounting brackets. The supervisor will 3D-print and hand over the sensor brackets, complete the final grading evaluation, and endorse the student's logbook records.
Supervisor Signature	

FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 14 / 08 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9
Student (Meeting Minute/ Achievements/ Activities)	The three 3D-printed HC-SR04 sensor mounting brackets were collected and installed on the parking diorama, replacing the temporary adhesive tape and improving cable management. The sensors were securely mounted and tested for accurate vehicle detection without false readings or movement. The final hardware-software integration was also verified, with parking occupancy updates syncing instantly across Firebase, the Flutter mobile app, and the React admin dashboard.
Supervisor (Suggestion & Comments)	The supervisor commended the tidy and professional appearance of the assembled diorama, particularly the custom mounting brackets and neatly concealed wiring.
Next Meeting Plan	Finalize the presentation slides & logbook and submit through LMS
Supervisor Signature	



FINAL YEAR PROJECT STUDENT LOGBOOK

LOGBOOK	
FYP Type	[] FYP 1 [✓] FYP 2
Date	: 17 / 08 / 2026 MEETING : 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10
Student (Meeting Minute/ Achievements/ Activities)	The comprehensive FYP 2 presentation slide was completed and submitted. The complete FYP 2 logbook was also compiled and finalized, documenting the FYP2 development, hardware, testing, and submission milestones.
Supervisor (Suggestion & Comments)	The supervisor acknowledged the presentation slide for review and the final logbook was also verified as complete, with all FYP 2 consultation milestones systematically recorded and ready for grading.
Next Meeting Plan	The FYP 2 project was officially completed, with all required deliverables, including the final thesis report, Innoverse poster, presentation slides, and logbook, verified, approved, and submitted. No further supervision meetings are required.
Supervisor Signature	